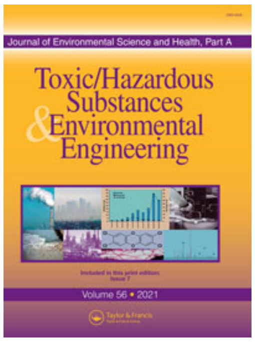

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Molecular detection of virulence markers to identify diarrhoeagenic *Escherichia coli* isolated from Mula-Mutha river, Pune District, India

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ABSTRACT

In this study presence of virulence genes in multidrug resistant *Escherichia coli* isolated from Mula-Mutha river, Pune, India was undertaken. The objective was to understand whether the isolates were of diarrhoeagenic or of environmental origin. This was essential since the river flows through urban and rural parts of Pune and its water is used not only for industrial and agricultural purposes but also for domestic usage. One hundred and two multidrug *E. coli* isolates were selected from our previous study which detected genes coding for antibiotic resistance as well as identified integrons associated with multidrug resistance. Isolates were subjected to multiplex PCR to detect presence of virulence genes, *set1A, set1B, sen astA, aggA, aafA, pet, stx1 and stx*. Sequencing was performed to confirm the amplified PCR product. Seven of the 102 *E. coli* isolates showed gene *set1A* alone identifying them as Enteroaggregative *E. coli*. Thus, the findings revealed that majority of drug resistant *E. coli* were environmental in origin. The presence of antibiotic resistant genes, integrons in the environment as well as diarrhoeagenic *E. coli* isolates is a warning and calls for efficient public health measures to ensure that untreated sewage and industrial waste does not enter the Mula-Mutha river.

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Antibiotic resistance; *E. coli*; Mula-Mutha; Shiga toxin; ShET 1A; environment; India

Introduction

The rapid emergence and spread of multidrug resistance organisms (MDRO) is a major global health concern.[1] This spread is alarming as it is not confined to clinical settings but in the community too, through the environment.[2,3] The presence of MDROs in contaminated water bodies increases the threat of community acquired infections since water bodies often serve as sources of drinking and other household activities. The discharge of untreated sewage and industrial waste into these water bodies is probably the contributing factor.[4] Detection of *Escherichia coli* and extended spectrum β-lactamases (ESBLs) from Indian rivers Ganges and Yamuna respectively and methicillin resistant *Staphylococcus aureus* (MRSA) in water bodies of Southern Austria are examples of local and global extent of this serious problem.[1,5,6]

River Mula-Mutha situated in Maharashtra State, India, has its origin in the Western Ghats and flows through urban (Pune city) and rural parts of Pune District. It has been documented that the river water is used not only for domestic practices but also for irrigation and industrial purposes.[7] It receives domestic, and hospital wastes as well as effluents from industries. At times it also receives agricultural run offs loaded with organic waste. Hence, like many other rivers, water in Mula-Mutha river is highly polluted and its water quality categorized as class IV.[8]

E. coli is a normal commensal of the gut of humans and animals. The presence of this bacterium in water bodies is linked with fecal contamination and considered as a primary indicator of drinking water quality.[9] However, *E. coli* strains can acquire virulence genes and cause different diseases including diarrhea.[10] The diarrhoeagenic *E. coli* (DEC) cause a spectrum of diarrhoeal illnesses, responsible for 30 to 40% of all the diarrheal episodes in developing countries.[11,12] These DEC have been characterized into six types namely, Enteropathogenic *E. coli* (EPEC); Enterotoxigenic *E. coli* (ETEC); Enterohemorrhagic *E. coli* (EHEC), Enteroaggregative *E. coli* (EAEC), the Enteroinvasive *E. coli* (EIEC) and the Diffusely adhering *E. coli* (DAEC).[10,13,14] The DEC can be differentiated by the presence of virulence genes in these strains.

Our earlier study had showed a high degree of fecal contamination in the Mula-Mutha River with a substantial proportion of the total coliform (TFC) isolated being multidrug resistant.[15,16] The presence of antibiotic resistant bacteria in river bodies is a cause of health concern.

As an extension of the above studies on Mula-Mutha, the present work was undertaken to study the virulence markers of *E. coli* strains with an aim to understand whether the isolates were of diarrhoeagenic or of environmental origin. A multiplex PCR was used to screen different genes so as to detect all categories of diarrheagenic *E. coli*.

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Table 1. List of primers and PCR product size of virulence genes analyzed in the study.

Sr. No.	Gene	Target	Primer	Size (bp)
1	*set1A*	Shet1A	F:TCACGCTACCATCAAAGA R:TATCCCCCTTTGGTGGTA	309
2	*set1B*	Shet1B	F:GTGAACCTGCTGCCGATATC R:ATTTGTGGATAAAAATGACG	147
3	*sen*	Shet2	F:ATGTGCCTGCTATTATTTAT R:CATAATAATAAGCGGTCAGC	799
4	*astA*	EAST1	F:ATGCCATCAACACAGTATAT R:GCGAGTGACGGCTTTGTAGT	110
5	*aggA*	AAF/I	F: TTAGTCTTCTATCTAGGG R: AAATTAATTCCGGCATGG	518
6	*aafA*	AAF/II	F: TGCGATTGCTACTTTATTAT R: ATTGACCGTGATTGGCTTCC	242
7	*pet*	Pet	F: ACTGGCGGACTCATTGCTGT R: GCGTTTTTCCGTTCCCTATT	832
8	*stx1*	Stx1	F: CAGTTAATGTGGTGGCGAAGG R: CACCAGACAATGTAACCGCTG	348
9	*stx2*	Stx2	F: ATCCTATTCCCGGGAGTTTACG R: GCGTCATCGTATACACAGGAGC	584

Materials and methods

Collection site and water sampling

Water samples were collected from Mula-Mutha river from different sites in and around Pune city over a period of one year (2016). Approximately 400 mL of river water was collected in sterilized 500 mL polypropylene bottles from 60 cm beneath the river surface, immediately kept on ice and transported to Mumbai (6–8 h) on the same day. The bottles were then stored overnight between 4–8 °C and processed further on next day.

Isolation of dual drug resistant (DDR) E. coli

The collected water samples were filtered through cellulose acetate membranes of 0.22 µm pore size. For isolation and enumeration of DDR thermotolerant fecal coliforms (TFC) the membranes were then placed on membrane fecal coliform (m-FC) agar in presence and absence of ceftazidime (16 µg/mL) and ciprofloxacin (4 µg/mL). The concentration of antibiotics was based on the guidelines given by the Clinical and Laboratory Standards Institute (CLSI, 2011).[17] Following a 24 h incubation at 44.5 °C, blue-coloured colonies appearing on the filter papers were enumerated. The concentration of DDR TFCs was calculated considering CFU per 100 mL in presence and absence of the two antibiotics incorporated. The DDR colonies from the antibiotic plate were then streaked on Hichrome *E. coli* agar (Himedia, Mumbai, India) to isolate DDR *E. coli* (bluish-green colonies). For further studies, DDR *E. coli* were preserved in sterile Luria Bertani (LB) broth (Himedia) supplemented with 15% glycerol and stored at −20 °C for further studies.

Antibiotic susceptibility testing (AST) and identification of multidrug resistant DDR E. coli strains

Frozen DDR *E. coli* were revived in LB and AST was undertaken by disk diffusion method using Mueller Hinton agar (Himedia) by the method described previously.[18] Hexadisc G15 minus (Himedia) with six antibiotics – ampicillin (10 µg), cefepime (30 µg), cefotaxime (30 µg), gentamicin (10 µg), imipenem (10 µg) and piperacillin/tazobactam (10/10 µg) were used. A pan sensitive strain (*E. coli* ATCC25922) was included as a negative control.[19] Antibiotic resistance was determined by measuring the respective zones of inhibition (diameters) around each disk.[17] This led to the identification of multidrug drug resistant DDR *E. coli* strains.

Positive controls for virulence study

Shigella flexneri strains from the characterized stock culture preserved in the Department of Microbiology, St. John's Medical College that were positive for *set1A*, *set1B* and *sen* by sanger sequencing were included as positive controls for these genes. For the remaining set of genes, *astA*, *aggA*, *aafA*, *pet*, *stx1* and *stx* EPEC B170 (serotype 0111:N4, obtained from Center of Disease Control, CDC, Atlanta), EPEC E2348 (serotype 0217: 46, a kind gift from Dr. S. Knutton, University of Birmingham, UK) and EIEC EI34 (serotype 0136:4, a kind gift from Dr. J. Nataro, Veterans Affairs Medical Center, Maryland, USA) were used as positive controls.

Detection of virulence genes

Virulence genes were detected using multiplex PCR. DNA was extracted from the isolates to screen for the presence of virulence genes mentioned above using the set of primers as listed in Table 1.[14,20,21] The included genes were selected so as to detect all categories of diarrheagenic *E. coli*.

PCR amplification was carried out using 5 μL of the DNA extract, 20 picomoles of primers and 25 μL of 2X DreamTaq master mix reagent in 50 μL reaction in a thermal cycler (Eppendorf). The PCR amplification was carried out with initial denaturation at 96 °C for 5 min followed by 35 cycles of 94 °C 30 sec, 57 °C for 30 sec and 72 °C for 1 min and a final step of 72 °C for 5 min.[21] Strains positive for the virulence genes were confirmed by sequencing using isolates with confirmed positivity. The amplified PCR products were analyzed using 1.5% agarose gel stained with ethidium bromide and visualized under UV light using a gel documentation system (Biorad).

Sequencing

The isolates that were PCR positive and showed bands on the gel, were outsourced to Medauxin, a Genomics R & D

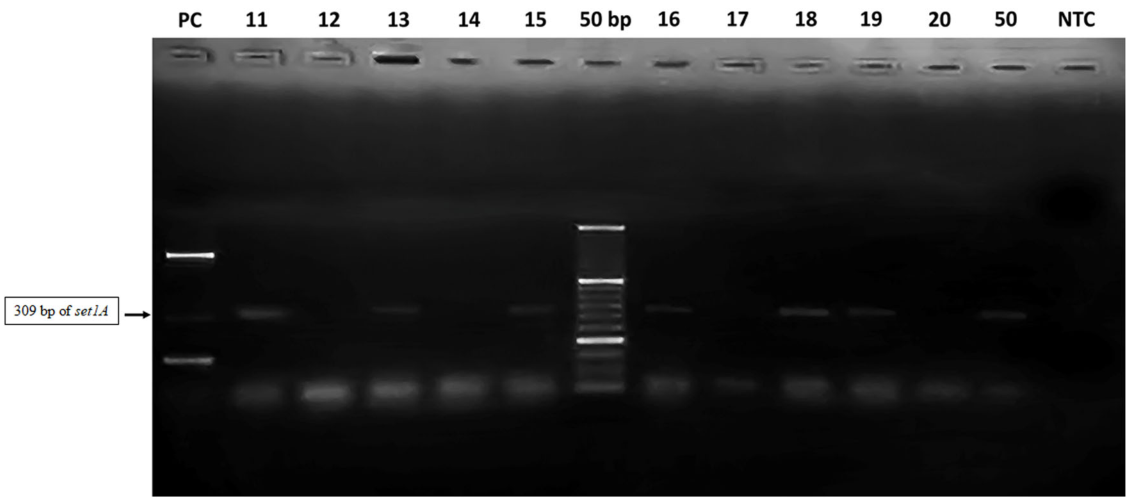

Figure 1. Agarose gel electrophoresis of *E. coli* isolates showing *shet1A* gene amplification (of 309 bp).
Key: PC-positive control, NTC- No template control, 50bp-DNA ladder.
Isolates positive for *set1A*: Lanes 11, 13, 15, 16, 18, 19 and 50
Isolates negative for *set1A*: Lanes 12, 14, 17, 20

Services company, Bengaluru, for sequencing for confirmation using the ABI PRISM Big Dye Terminator cycle sequencing ready reaction kit (Applied Biosystems, CA) in an automated DNA sequencer (ABI PRISM 310 Genetic Analyzer). The obtained sequences were aligned using ClustalW program in MEGA6 software, and the aligned sequences were compared with the published sequences available from GenBank database using NCBI's BLAST search (http://www.ncbi.nlm.nih.gov/BLAST/).

Results

DDR E. coli strains used in the present study

A total of 219 DDR *E. coli* strains resistant to ceftazidime and ciprofloxacin were isolated from Mula-Mutha river. Based on AST, 102 DDR strains which were resistant to additional four or more antibiotics on the hexadisc were selected for the present work. Thus, the study shortlisted *E. coli* strains resistant to minimum six antibiotics (ceftazidime and ciprofloxacin being initially used for screening of DDR).

Presence of virulence genes

A total of 102 strains were tested. Only seven isolates showed positive bands (Figure 1) for *set1A* thus identifying them as EAEC. Presence of *set1A* gene was confirmed in four of the seven positive samples through sequencing. No other isolate showed positive bands for any other primers.

Discussion

Several factors contribute towards pollution of the river Mula-Mutha. The river passes through the city of Pune and is the major source of water for villages along its bank. It receives domestic and industrial sewage as well as hospital waste which is not always adequately treated.[22,23] The presence of organic pollutants as well as heavy metals leads to an increase in AMR and inorganic compounds (calcium) in the environment are known to promote gene transfer by altering the uptake capacity of microbes.[24] Sadly, the environmental dimension of the emergence and spread of resistance is often neglected.[25] The scoping report on the AMR research landscape in India has highlighted high levels of antibiotic resistant bacteria and antibiotic resistant genes in various water bodies.[26]

A high degree of fecal contamination in the river water of Mula-Mutha was documented in our previous study. Further using a combination of two antibiotics ceftazidime and ciprofloxacin, presence of 219 drug resistant *E. coli* was reported.[15] The selection of these two antibiotics was not only based on literature[27,28] but also combined with the information collected on the sale of related to antibiotics over the counter and based on the commonly prescribed antibiotics by the Public Health Center (PHC) doctors or private practitioners in selected regions around the study area (data not shown). Thus, the existence of DDR *E. coli* in Mula-Mutha river reflects the heavy burden of resistant pathogens posing threat to community members relying on the river for their water requirements. Further studies were then directed toward detection of genes responsible in these DDR isolates for β-lactamase production *tem*, *shv*, *ctxm15* and *ctxm27*, quinolone resistance (*qnrA*, *qnrB*, *qnrS*, *oqxA*) and possible vectors for horizontal gene transfer genes represented by Class 1, 2 integrons (*intI1*, *intI2*).[16]

In the present study 102 DDR *E. coli* isolates resistant to additional four or more antibiotics were selected to determine if these isolates were diarrhoeagenic. Contrary to our expectations, *Shigella* Enterotoxin gene (*set1A*) was detected in only seven isolates characterizing these isolates as EAEC which is an important cause of traveller's diarrhea in many regions.[14,29] Other reports on DEC from Bangladesh and Egypt express concern over the presence of DEC in surface waters in urban areas with EPEC, EHEC, ETEC being more common than EAEC and EIEC.[30–32]

Correlating the earlier data on antibiotic associated genes,[16] it was noted that six of the seven *set1A* positive strains had *ctxm-15* gene, which is commonly reported by Indian researchers.[33,34] Although all of these seven strains were resistant to ciprofloxacin, plasmid mediated resistance to quinolones (*qnr* genes) was detected in only one isolate probably because resistance to quinolones is mediated more commonly through chromosomes.[35] This implies that horizontal gene transfer for quinolones was minimal. Likewise, it was seen that Class 1 integron was present in three of the seven *set1A* positive isolates. This is noteworthy as Class 1 integron-positive isolates are reported to be more frequently resistant to multiple classes of antibiotics than integron-negative isolates. Moreover, they are also associated with spread of antibiotic resistance in Gram negative bacilli in the community.[36,37] The *set1A* positivity in multidrug resistant *E. coli* combined with presence of integron Class 1 from a flowing water source as seen in this study should be taken as a warning.

Nevertheless, our findings suggest that majority of multidrug resistant *E. coli* tested are environmental in origin. Several other studies have also documented drug resistant pathogens in polluted rivers. For example, multidrug resistant *E. coli* along with other bacteria like *Morganella morganii*, *Aeromonas hydrophila*, *Providencia alcalifaciens*, *Proteus mirabilis* as well as diarrhoeagenic pathogens belonging to *Shigella* sp., were reported to be present in Salmon river, British Columbia.[36–38] In another study, multidrug resistant *Klebsiella* spp. were documented from river Yamuna, New Delhi, India.[39]

The environmental exposure of antimicrobials is linked to development of multidrug resistance.[40] This is compounded by the overuse, misuse and indiscriminate use of antibiotics for humans, veterinary as well as aquaculture use.[41] Moreover, India does not have a buy back drug policy and/or efficient treatment of effluent from pharmaceutical manufacturing industries.[42] Hence, the detection of drug resistant environmental *E. coli* in the present study could be due to the increasing levels of antibiotics in the environment including discarded antibiotics. However, the presence of chemicals, other than antibiotics, such as those present in disinfectants and/or their by products, nano materials or even heavy metals which have been documented to stimulate antibiotic resistance cannot be ruled out.[43–48]

Horizontal transfer of mobile genetic elements either through integrons, plasmids or transposons has been associated with spread and acquisition of antibiotic resistance.[49,50] As documented from our earlier study integrons were present in 66% of these isolates.[16] Thus, it is likely that the mobile genetic elements from these *E. coli* could be transferred to other pathogenic organisms.

Hence, more efficient public health measures need to be in place to ensure that untreated sewage does not enter water bodies. Appropriate treatment strategies to prevent or reduce entry of untreated water, domestic pollutants (eg. chemical detergents), organic pollutants, fertilizers, heavy metals into water bodies should be strongly implemented.[51,52] Additionally, it may also be worth considering the recommendations related to unsegregated waste disposal put forth to reduce the threat of antimicrobials in ground and surface water.[53] These include the implementation of a policy for the separation and destruction of bioactive compounds in drugs so as to circumvent their accumulation in the environment, use of an appropriate surveillance system for the constant monitoring of these compounds in water bodies and generating awareness amongst general public toward the disposal of active pharmaceutical compounds (APCs). All efforts together would in turn ensure a healthier ecosystem as well as prevent outbreaks of diseases like diarrhea and dysentery in the communities that use this water.

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Disclosure statement

The authors declare that they have no potential conflict of interest.

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